1. INTRODUCTION

1.1 Project Overview

Electricity plays a crucial role in economic development, industrial growth, and improvement of living standards. Monitoring electricity consumption helps governments and policymakers understand demand patterns, infrastructure requirements, and economic activity across regions.

India has relatively low electricity tariffs, but its **per capita electricity consumption remains lower than the global average**, indicating the potential for increased demand and expansion of energy infrastructure.

The **COVID-19 pandemic in 2020 significantly affected electricity consumption patterns** due to lockdowns, restrictions on industries, and increased residential energy usage.

This project analyses **state-wise electricity consumption data across India from January 2019 to December 2020**, with special focus on:

- Pre-pandemic electricity trends
- Impact of COVID-19 lockdown
- Regional consumption variations
- Post-lockdown recovery patterns

The analysis is conducted using **SQL for data management, Tableau for visualization, and Flask for web integration**.

1.2 Purpose

The main purpose of this project is to analyse electricity consumption patterns across Indian states and identify the impact of COVID-19 lockdown on energy usage.

The objectives include:

- Understanding **overall electricity consumption trends**
- Identifying **regional demand differences**
- Studying **recovery of electricity usage after lockdown**
- Creating **interactive dashboards using Tableau**
- Integrating dashboards into a **web interface using Flask**

2. IDEATION PHASE

2.1 Problem Statement

Electricity demand fluctuates due to multiple factors such as economic activity, seasonal variations, and unexpected events like pandemics. The COVID-19 lockdown significantly disrupted industrial production and mobility, which likely influenced electricity consumption across Indian states.

However, there is limited consolidated analysis that clearly visualizes:

- State-wise electricity demand changes
- Regional electricity consumption trends
- Recovery patterns after lockdown

Therefore, this project aims to **analyse electricity consumption data and provide visual insights into these changes.**

2.2 Empathy Map Canvas

Stakeholders include:

- Government energy departments
- Electricity distribution companies
- Researchers
- Policy makers

Think	Feel
Need accurate energy demand forecasts	Concern about energy shortages
Want efficient resource planning	Interested in recovery trends

Say	Do
"Electricity demand dropped during lockdown."	Monitor consumption statistics
"Energy demand is recovering slowly."	Analyse state-wise reports

2.3 Brainstorming

Possible analytical ideas considered:

- Monthly electricity consumption trends
- Comparison between **2019 and 2020 consumption**
- Impact of lockdown months
- Regional electricity demand comparison
- Top electricity consuming states
- Recovery rate after lockdown

Final selected scenarios:

1. Overall consumption trend changes
2. Regional electricity demand variation
3. Recovery after lockdown

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System Response
Data Input	Dataset uploaded	Data stored in SQL database
Data Processing	SQL queries executed	Clean dataset prepared
Visualization	Data sent to Tableau	Charts and dashboards created
Analysis	User interacts with dashboard	Filters and insights generated
Web Access	Dashboard embedded	User accesses visualization via web

3.2 Solution Requirement

Functional Requirements

- Import electricity consumption dataset
- Store data in SQL database
- Perform SQL queries
- Create Tableau dashboards
- Implement filters and calculated fields
- Embed dashboard in Flask web interface

Non-Functional Requirements

- Fast dashboard loading
- Interactive visualizations
- Responsive web interface
- Scalable database structure

3.3 Data Flow Diagram

Level 0 DFD

User → Data Input → SQL Database → Tableau Visualization → Web Interface

Level 1 DFD

Dataset
↓
Data Cleaning
↓
SQL Database
↓

Data Processing Queries
↓
Tableau Visualization
↓
Dashboard & Story
↓
Flask Web Application

3.4 Technology Stack

Technology	Purpose
SQL (MySQL/PostgreSQL)	Data storage and querying
Tableau	Data visualization and dashboard creation
Python Flask	Web application integration
HTML/CSS	Frontend UI
Dataset	Electricity consumption data (2019–2020)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed system solves the problem of understanding electricity consumption patterns by transforming raw electricity usage data into meaningful visual insights through interactive dashboards.

This allows users to easily identify trends, compare regions, and analyze recovery after lockdown.

4.2 Proposed Solution

The solution involves:

1. Collecting electricity consumption dataset
2. Storing the dataset in SQL database
3. Performing data cleaning and preprocessing
4. Connecting SQL database with Tableau
5. Creating visual dashboards and stories
6. Integrating dashboards into a web interface using Flask

4.3 Solution Architecture

Dataset
↓

SQL Database
↓
Data Processing (SQL Queries)
↓
Tableau Visualization
↓
Interactive Dashboard
↓
Flask Web Application
↓
User Interface

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity
Phase 1	Problem Definition
Phase 2	Data Collection
Phase 3	Database Creation
Phase 4	Data Cleaning & Preparation
Phase 5	Tableau Visualization
Phase 6	Dashboard & Story Creation
Phase 7	Web Integration using Flask
Phase 8	Testing and Performance Evaluation

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance evaluation is based on:

- Amount of data rendered in database
- Speed of dashboard loading
- Efficiency of filters and calculations

Performance metrics include:

- Data rendering speed
- Query execution time
- Dashboard response time

7. RESULTS

7.1 Output Screenshots

The project generates several visual outputs such as:

- Monthly electricity consumption trend
- Regional electricity demand comparison
- Lockdown impact analysis
- State-wise electricity consumption

Screenshots of dashboards and graphs are included in this section.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides clear visualization of electricity trends
- Helps policymakers analyze energy demand
- Identifies impact of COVID-19 on electricity usage
- Interactive dashboard improves data exploration

Disadvantages

- Limited dataset time period
- Does not include future predictions
- Dependent on dataset accuracy

9. CONCLUSION

The project successfully analyses electricity consumption patterns across Indian states between 2019 and 2020. The analysis highlights the **significant impact of COVID-19 lockdown on electricity demand**, showing a sharp decline followed by gradual recovery.

Through the use of **SQL, Tableau, and Flask**, the project demonstrates how data analytics and visualization tools can provide meaningful insights into energy consumption trends.

10. FUTURE SCOPE

Future improvements may include:

- Adding **machine learning models for electricity demand prediction**
- Including **more recent datasets (2021–2025)**
- Real-time electricity monitoring dashboards
- Mobile-friendly dashboard interface

- Integration with government energy data portals

11. APPENDIX

Source Code

Contains SQL queries, Tableau workbook, and Flask integration code.

Example SQL Query:

```
SELECT state, SUM(consumption)
FROM electricity_data
GROUP BY state;
```

GitHub & Project Demo Link

GitHub Repository:
https://github.com/0303shreyas/Smart-Bridge.git